

Ivalin Ivanov

DevOps • DevSecOps focus | Certified Kubernetes Administrator (CKA)

Targovishte, Bulgaria • Open to remote • ive.mcfire@gmail.com

github.com/ivemcfire • ivemcfire.github.io • linkedin.com/in/ivemcfire

Professional Summary

Informatics graduate and returning IT professional moving into DevOps with a security lean. Started as a Windows sysadmin at Siemens IT Solutions and Services in the US, then networking and support work in Australia. Since 2016 the sole technical person in a family IoT and IP-camera systems business with 200+ customers. For a year and a half I have focused on DevOps study and practice: the five-module SoftUni DevOps and Cloud program, the KubeCraft Career Accelerator and the CKA certification, with the CKS planned. I run a self-built highly available Kubernetes cluster with always-on workloads and publish post-mortems of its real incidents. Now building depth in AWS, Terraform and CI/CD.

Professional Certifications

- **Certified Kubernetes Administrator (CKA)** — The Linux Foundation, 2026
- **SoftUni module certificates** — Linux System Administration (92%), Containerization and Infrastructure (100%), CI/CD & Monitoring (99%), 2026. Kubernetes module certificate expected November 2026
- **MCSE** (with MCSA and MCP) — Microsoft, 2005–2006

Technical Skills

Kubernetes: k3s, embedded etcd HA, Helm, MetalLB, Traefik, NVIDIA device plugin, RBAC and NetworkPolicy (CKA)

Linux: Ubuntu, Debian, systemd, journalctl, netplan, udev, Bash; performance debugging with pidstat, perf and pprof

Containers and CI/CD: Docker, containerd, buildx multi-arch builds (amd64 + arm64), GitHub Actions, GHCR, Gitea registry; Jenkins (training)

Infrastructure as Code: Terraform, Ansible (SoftUni, fundamentals and advanced); AWS in progress

Observability: Prometheus, Alertmanager, Grafana, Loki, Uptime Kuma

Networking and security: Cloudflare Tunnel and Access SSO, Tailscale / WireGuard, SSH hardening, fail2ban, age-encrypted secrets, IoT egress isolation, Cisco routers

Legacy: Windows Server, Active Directory, Exchange

Homelab Projects

Self-Hosted HA Kubernetes Platform — [homelab](#)

Jan 2026 – present

- 5-node k3s cluster: 2 x86 servers and 3 arm64 OnePlus phones on postmarketOS. 3-member etcd quorum using an etcd-only member; failover tested by stopping a server while the API stayed up. About 50 pods across 15 namespaces.
- No inbound ports: public services through Cloudflare Tunnel (2 replicas) behind Access SSO; administration over a Tailscale mesh with redundant subnet routers.
- Monitoring outside the failure domain: Prometheus, Alertmanager and Loki in-cluster, Grafana and Uptime Kuma on a separate host, alerts by SMTP, routing tuned to actionable alerts only.
- Profiled the control plane in production through the kubelet pprof proxy and found a leaked etcd goroutine; the upgrade cut k3s-server CPU from 104% to about 10%.
- Found a backup CronJob that had not completed once in 28 days; rebuilt it with dependencies baked into the image and added job-failure alerting.
- GitHub Actions builds multi-arch images to GHCR; every manifest kept in Git; incidents published as post-mortems. Built with heavy AI assistance (Claude Opus), every change verified on the hardware.

Self-Hosted Video Surveillance on Kubernetes — [homelab](#)

2026 – present

- Frigate NVR on 6 IP cameras with TensorRT object detection on a GTX 1050 Ti, scheduled through the NVIDIA device plugin. CPU and memory limits plus kubelet reservations ended repeated node lockups.
- Cameras have no route off the LAN; remote viewing only through SSO-gated Cloudflare Access.

Work Experience

Systems & Network Engineer — [Vividea LTD, Bulgaria](#)

2016 – present

- Sole technical person in a family-owned IT and security systems company: 200+ customers and 200+ installations across homes, shops and offices.
- PC and server support for small businesses, network setup and on-site troubleshooting.

- IP camera systems with local hosting on Blue Iris and Hikvision; next generation (Frigate on Kubernetes, ANNKE cameras, GPU detection) tested in the homelab.

IT Support Technician — [LowCost IT, Brisbane, Australia](#) **2010 – 2012**

- Network setup and Cisco router configuration, hardware repairs and customer support.

Windows System Administrator — [Siemens IT Solutions and Services, Illinois, USA](#) **2007 – 2009**

- Supported client sites Siemens Energy & Automation (Elk Grove) and KPMG (Schaumburg): Windows servers, Active Directory, Exchange mailboxes and backups.
- On-call for after-hours incidents, including a Friday 9 PM server-room failure at a remote site affecting 500+ users.

Education and Training

DevOps and Cloud 2025 — [SoftUni, Bulgaria](#) **2025 – 2026**

Linux System Administration · Containerization and Infrastructure (Docker, Terraform, Ansible) · Kubernetes · CI/CD & Monitoring (GitHub Actions, Jenkins, GitOps) · Azure Essentials

KubeCraft Career Accelerator — [hands-on Kubernetes career program](#) **Mar 2026 – Mar 2027**

Homelab-first program. Following its content from January 2026 started my homelab; joined the program in March 2026.

MCSE Certification Program — [NetXperts, Inc., Des Plaines, IL, USA](#) **2005 – 2006**

Microsoft MCP, MCSA and MCSE; completed the Cisco CCNA course (no exam).

Bachelor's in Informatics and Computer Science — [University of Economics – Varna, Bulgaria](#) **1999 – 2004**

Spoken Languages

Bulgarian — [native](#) · **English** — [fluent](#), professional use in the US and Australia